

INTRODUCTION TO APPLIED NUMERICAL METHODS

Curve Fitting: Linear Regression

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LEAST-SQUARES FIT OF A STRAIGHT LINE

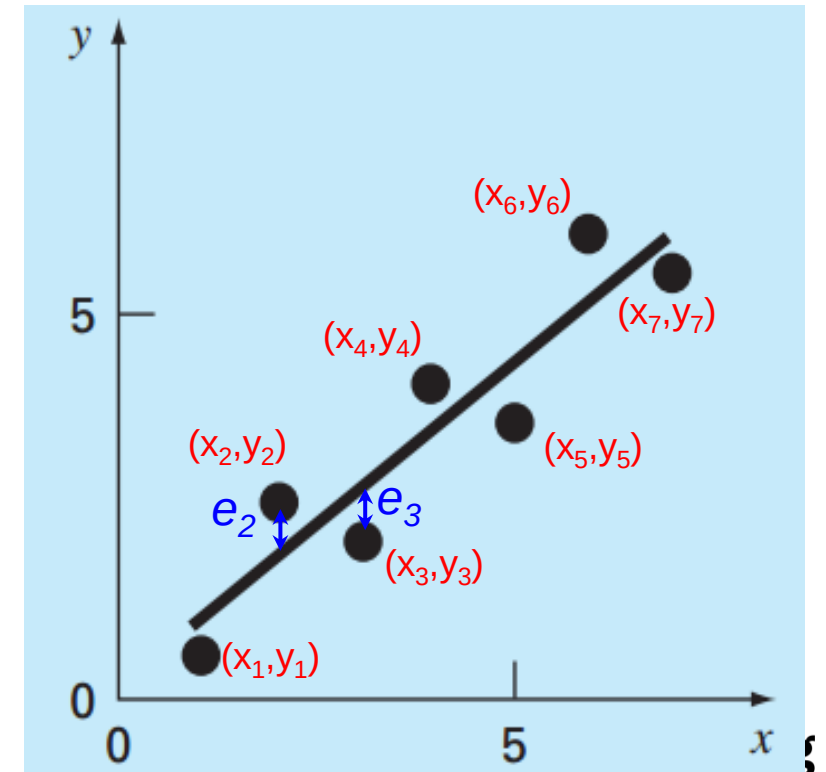
$$S_r = \sum_{i=1}^n e_i^2 = \sum_{i=1}^n (y_{i,\text{measured}} - y_{i,\text{model}})^2 = \sum_{i=1}^n (y_i - a_0 - a_1 x_i)^2$$

$$\frac{\partial S_r}{\partial a_0} = -2 \sum (y_i - a_0 - a_1 x_i) = 0$$

$$\frac{\partial S_r}{\partial a_1} = -2 \sum [(y_i - a_0 - a_1 x_i) x_i] = 0$$

$$a_1 = \frac{n \sum x_i y_i - \sum x_i \sum y_i}{n \sum x_i^2 - (\sum x_i)^2}$$

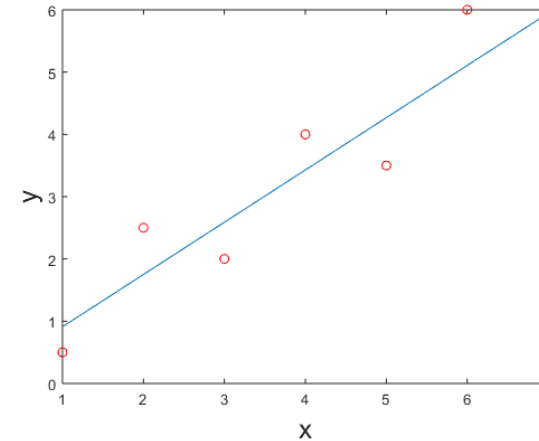
$$a_0 = \bar{y} - a_1 \bar{x}$$



AN EXAMPLE

Linear regression of the x and y values in the first 2 columns in the Table

x_i^2	$x_i y_i$	x_i	y_i
1	0.5000	1	0.5
4	5.0000	2	2.5
9	6.0000	3	2.0
16	16.0000	4	4.0
25	17.5000	5	3.5
36	36.0000	6	6.0
49	38.5000	7	5.5
Σ		Σ	24.0



Solution:

First necessary quantities for linear regression will be computed:

$$n = 7 \quad \sum x_i y_i = 119.5 \quad \sum x_i^2 = 140$$

$$\sum x_i = 28 \quad \bar{x} = \frac{28}{7} = 4$$

$$\sum y_i = 24 \quad \bar{y} = \frac{24}{7} = 3.428571$$

$$a_1 = \frac{n \sum x_i y_i - \sum x_i \sum y_i}{n \sum x_i^2 - (\sum x_i)^2}$$

$$a_0 = \bar{y} - a_1 \bar{x}$$

$$a_1 = \frac{7(119.5) - 28(24)}{7(140) - (28)^2} = 0.8392857$$

$$a_0 = 3.428571 - 0.8392857(4) = 0.07142857$$

$$\Rightarrow y = 0.07142857 + 0.8392857x$$

ERROR OF LINEAR REGRESSION

Sum of the Squared Error

$$S_r = \sum_{i=1}^n e_i^2 = \sum_{i=1}^n (y_i - a_0 - a_1 x_i)^2$$

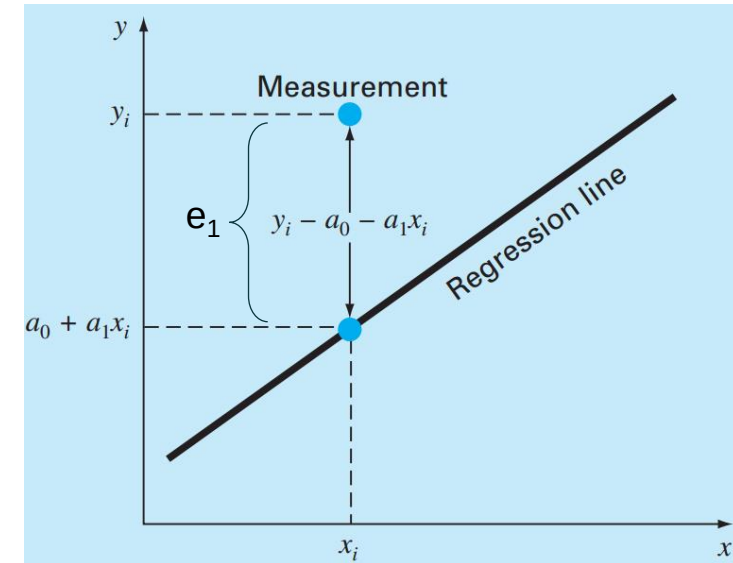
Standard Error of the Estimate ($s_{y/x}$)

$$s_{y/x} = \sqrt{\frac{S_r}{n-2}}$$

← Degree of freedom (for data used in linear regression)

$s_{y/x}$: Quantifies the spread around the regression line; **but it does not provide a quantitative and relative measure of “goodness of fit”**

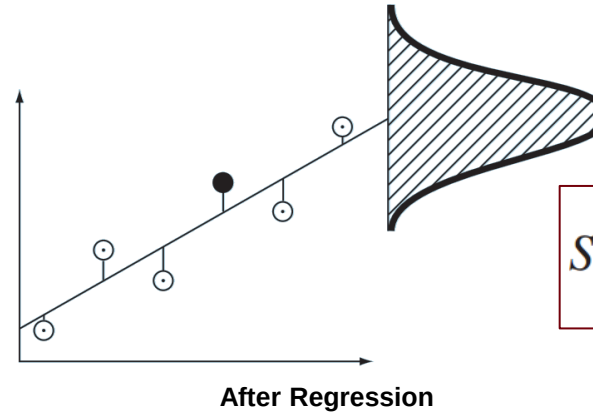
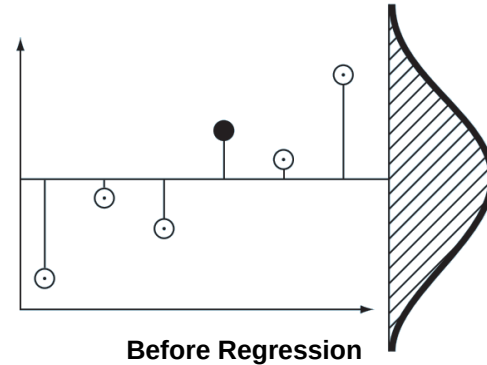
y/x notation designates the error is for a predicted value of y corresponding to a particular x



ERROR OF LINEAR REGRESSION

Coefficient of Determination

$$S_t = \sum (y_i - \bar{y})^2$$



$$S_r = \sum_{i=1}^n e_i^2 = \sum_{i=1}^n (y_i - a_0 - a_1 x_i)^2$$

$$r^2 = \frac{S_t - S_r}{S_t}$$

Improvement or error reduction
due to the linear regression

Error before the linear regression

The coefficient of determination

r : correlation coefficient $r = \sqrt{\frac{S_t - S_r}{S_t}}$

$S_r=0$ and $r = r^2 = 1$: the line explains 100% of the variability of the data (perfect fit)

$S_r = S_t$ and $r = r^2 = 0$: regression yields no improvement (poor fit)

AN EXAMPLE

Compute the total standard deviation (s_y), the standard error of the estimate ($s_{y/x}$), and the correlation coefficient (r) for the data below

Solution:

Standard Deviation (s_y)

$$s_y = \sqrt{\frac{S_t}{n-1}} = \sqrt{\frac{\sum (y_i - \bar{y})^2}{n-1}} = \sqrt{\frac{22.7143}{6}} = 1.9457$$

Error in Textbook

x_i	y_i	$(y_i - \bar{y})^2$	$(y_i - a_0 - a_1 x_i)^2$
1	0.5	8.5765	0.1687
2	2.5	0.8622	0.5625
3	2.0	2.0408	0.3473
4	4.0	0.3265	0.3265
5	3.5	0.0051	0.5896
6	6.0	6.6122	0.7972
7	5.5	4.2908	0.1993
Σ	24.0	22.7143	2.9911

Standard Error of the Estimate ($s_{y/x}$)

$$S_r = \sum_{i=1}^n e_i^2 = \sum_{i=1}^n (y_i - a_0 - a_1 x_i)^2 \Rightarrow s_{y/x} = \sqrt{\frac{S_r}{n-2}} = \sqrt{\frac{2.911}{5}} = 0.7735$$

Correlation coefficient (r)

$$r^2 = \frac{S_t - S_r}{S_t} = \frac{22.7143 - 2.911}{22.7143} = 0.868 \Rightarrow r = \sqrt{\frac{S_t - S_r}{S_t}} = \frac{22.7143 - 2.911}{22.7143} = 0.932$$

LINEARIZATION OF NONLINEAR RELATIONSHIPS

- Nonlinear model for simple power equation
 - $y = ax^b$
- Transform the equation to linear expression : e.g. take $\ln$ at both sides
 - $\ln(y) = \ln(ax^b) = \ln(a) + b \ln(x)$
 - Linear equation for $\ln(y) \sim \ln(x)$

x	y	Ln(x)	Ln(y)
1	0.5	0	-0.301
2	1.7	0.301	0.226
3	3.4	0.477	0.534
4	5.7	0.602	0.753
5	8.4	0.699	0.922

- $y = ae^{bx} \rightarrow \ln(y) = \ln(a) + bx$

- $y = \frac{ax}{b+x} \rightarrow \frac{1}{y} = \frac{b+x}{ax} = \frac{1}{a} + \frac{b}{a} \frac{1}{x}$

